Functional Constraints

Flame competition constraint within pocket regions.

Penalty coefficient 0.0100

Maximum allowable heat flux density ratio (max/min) 100.00

The linear burning rate of the first conditional fuel is greater than that of the second.

Penalty coefficient 0.0100

Constraint on maximum kinetic flame heat flux density.

Penalty coefficient 0.0100

Maximum kinetic flame heat flux density in inter-pocket medium 1.00e+09

Maximum kinetic flame heat flux density in skeleton structure 1.00e+08

Maximum kinetic flame heat flux density in pocket region outside skeleton 1.00e+08

Constraint on skeleton thickness to pore diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to pore diameter ratio 3.00

Constraint on skeleton thickness to large oxidizer particle diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to large oxidizer particle diameter ratio 1.00

Parametric Constraints

Constraint on allowable surface temperature range

Minimum surface temperature 600 K

Maximum surface temperature 750 K

Differential Evolution Algorithm

Population size: 384

Mutation force: 0.5

Crossover probability: 0.9

Number of threads: 32

Termination strategy: OrTerminationStrategy

Group Combustion Solver Parameters

Shared Parameters (All Compositions)

Pre-exponential coefficient for gas phase reactions in inter-pocket medium 358533657878.7982

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas phase reactions in inter-pocket medium 249995.41941130697

Parameter bounds [5.000E+004; 2.500E+005]

Pre-exponential factor for gas-phase reactions in pocket region outside skeleton 1208262.0805067045

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions in pocket region outside skeleton 50000.002751819455

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in inter-pocket medium 2.499998657764581

Parameter bounds [0.000E+000; 2.500E+000]

Reaction order for gas-phase chemistry outside skeleton structure 1.4642763863821644

Parameter bounds [0.000E+000; 2.500E+000]

Enthalpy difference between vaporization and binder decomposition 847610.8404392304

Parameter bounds [-1.000E+007; 1.000E+007]

Diffusion flame height correlation coefficient 0.0967

Parameter bounds [1.000E-003; 1.000E+001]

Power law exponent for burning rate dependency on A (skeleton thickness) 1.0000

Parameter bounds [1.000E+000; 1.000E+000]

Power law exponent for burning rate dependency on B (pore diameter) 2.0000

Parameter bounds [2.000E+000; 2.000E+000]

Radiation temperature correlation coefficient (average layer temperature) 0.9980

Parameter bounds [0.000E+000; 1.000E+000]

Bas_2 (includes Bas_3, Bas_4) - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 999988600.7989916

Parameter bounds [1.000E+000; 1.000E+009]

Activation energy for condensed phase reactions 95906.06551654736

Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 9190375350377.504
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 212440.73472615622
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 1.0314643416197686
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 4.604985970704636E-07
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 1.0004768417534463E-12
Parameter bounds [1.000E-012; 1.000E-005]

Bas_1 - Condensed Phase Parameters
Pre-exponential coefficient for condensed phase reactions 999981676.1306888
Parameter bounds [1.000E+000; 1.000E+009]
Activation energy for condensed phase reactions 89150.02558971233
Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 7486479906474.34
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 157207.42562871464
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 1.4715684223857184
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 6.575795130362058E-08
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 1.1661128500571199E-12
Parameter bounds [1.000E-012; 1.000E-005]

Bas_0 - Condensed Phase Parameters
Pre-exponential coefficient for condensed phase reactions 999981908.2654762
Parameter bounds [1.000E+000; 1.000E+009]
Activation energy for condensed phase reactions 95906.00597253634
Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 108113716.55200002
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 50000.0506384706
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 0.5339646868310195
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 5.729062578733727E-07
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 1.0637030116358129E-12
Parameter bounds [1.000E-012; 1.000E-005]

Optimization Results
Objective function value 0.2312
Total penalty from constraint violations 0.0000

Propellant Composition "Bas_2"

At pressure 1 MPa
Experimental burning rate 16.94 mm/s
Calculated burning rate 9.49 mm/s
First Conditional Fuel
Surface temperature 648.81 K
Linear burning rate 11.46 mm/s
Mass decomposition rate 19.01 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 2.61e+07 W/m^2

Kinetic flame height 11.71 μm
Average kinetic flame density 1.82 kg/m^3
Average kinetic flame temperature 2,177.91 K
Binder sublimation heat flux density 2.61e+07 W/m^2
Surface heat balance error -0.005 W/m^2

Secondary Conditional Fuel

Surface temperature 616.62 K
Linear burning rate 4.53 mm/s
Mass decomposition rate 7.51 $kg{\cdot}s^{-1}{\cdot}m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 2.09e+06 W/m^2
Kinetic flame height 83.73 μm
Average kinetic flame density 2.66 kg/m^3
Average kinetic flame temperature 1,490.81 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.2e+07 W/m^2
Kinetic flame height 8.15 μm
Average kinetic flame density 3.6 kg/m^3
Average kinetic flame temperature 1,103.81 K
Heat flux density due to metal combustion, 2.18e+06 W/m^2
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 4.91e-06 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.26e-04 m
Pore diameter 5.18e-08 m

Diffusion flame heat flux density 3.55e+06 W/m^2
Diffusion flame height 84.17 μm
Heat flux density to surface within skeletal structure 5.07e+06 W/m^2
Heat flux density to surface outside skeletal structure 1.34e+06 W/m^2
Total heat flux density to surface 9.96e+06 W/m^2
Binder sublimation heat flux density 9.96e+06 W/m^2
Surface heat balance error 0.0012 W/m^2

At pressure 4.06 MPa

Experimental burning rate 35.09 mm/s
Calculated burning rate 16.57 mm/s

First Conditional Fuel

Surface temperature 721.57 K
Linear burning rate 68.8 mm/s
Mass decomposition rate 114.14 $kg{\cdot}s^{-1}{\cdot}m^{-2}$
Kinetic flame heat flux density to surface 1.69e+08 W/m^2
Kinetic flame height 1.76 μm
Average kinetic flame density 7.27 kg/m^3
Average kinetic flame temperature 2,214.28 K
Binder sublimation heat flux density 1.69e+08 W/m^2
Surface heat balance error 0.024 W/m^2

Secondary Conditional Fuel

Surface temperature 631.63 K
Linear burning rate 7.06 mm/s
Mass decomposition rate 11.72 $kg{\cdot}s^{-1}{\cdot}m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.04e+07 W/m^2
Kinetic flame height 16.6 μm
Average kinetic flame density 10.74 kg/m^3
Average kinetic flame temperature 1,498.32 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.65e+07 W/m^2
Kinetic flame height 5.82 μm

Average kinetic flame density 14.48 kg/m^3
Average kinetic flame temperature 1,111.32 K
Heat flux density due to metal combustion, 3.33e+06 W/m^2
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 2.17e-06 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 8.11e-05 m
Pore diameter 2.13e-08 m

Diffusion flame heat flux density 2.26e+06 W/m^2
Diffusion flame height 131.28 µm
Heat flux density to surface within skeletal structure 6.54e+06 W/m^2
Heat flux density to surface outside skeletal structure 6.99e+06 W/m^2
Total heat flux density to surface 1.58e+07 W/m^2
Binder sublimation heat flux density 1.58e+07 W/m^2
Surface heat balance error -4.65e-04 W/m^2

At pressure 6.5 MPa

Experimental burning rate 44.85 mm/s
Calculated burning rate 20.9 mm/s

First Conditional Fuel

Surface temperature 750 K
Linear burning rate 126.12 mm/s
Mass decomposition rate 209.23 kg·s^{-1}·m^{-2}
Kinetic flame heat flux density to surface 3.19e+08 W/m^2
Kinetic flame height 0.93 µm
Average kinetic flame density 11.58 kg/m^3
Average kinetic flame temperature 2,228.5 K
Binder sublimation heat flux density 3.19e+08 W/m^2
Surface heat balance error 0.053 W/m^2

Secondary Conditional Fuel

Surface temperature 639.27 K
Linear burning rate 8.79 mm/s
Mass decomposition rate 14.58 kg·s^{-1}·m^{-2}

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.68e+07 W/m^2
Kinetic flame height 10.28 µm
Average kinetic flame density 17.17 kg/m^3
Average kinetic flame temperature 1,502.14 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.72e+07 W/m^2
Kinetic flame height 5.54 µm
Average kinetic flame density 23.13 kg/m^3
Average kinetic flame temperature 1,115.14 K
Heat flux density due to metal combustion, 4.09e+06 W/m^2
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 1.45e-06 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 6.52e-05 m
Pore diameter 1.38e-08 m

Diffusion flame heat flux density 1.82e+06 W/m^2
Diffusion flame height 163.31 µm
Heat flux density to surface within skeletal structure 5.78e+06 W/m^2
Heat flux density to surface outside skeletal structure 1.22e+07 W/m^2
Total heat flux density to surface 1.98e+07 W/m^2
Binder sublimation heat flux density 1.98e+07 W/m^2
Surface heat balance error -0.0023 W/m^2

Propellant Composition "Bas_3"

<u>*At pressure 1 MPa*</u>
Experimental burning rate 15.24 mm/s
Calculated burning rate 15.3 mm/s
First Conditional Fuel
Surface temperature 648.81 K
Linear burning rate 11.46 mm/s
Mass decomposition rate 19.01 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 2.61e+07 W/m^2
Kinetic flame height 11.71 µm
Average kinetic flame density 1.82 kg/m^3
Average kinetic flame temperature 2,177.91 K
Binder sublimation heat flux density 2.61e+07 W/m^2
Surface heat balance error -0.005 W/m^2
Secondary Conditional Fuel
Surface temperature 648.81 K
Linear burning rate 11.46 mm/s
Mass decomposition rate 19.01 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 832,702.72 W/m^2
Kinetic flame height 206.1 µm
Average kinetic flame density 2.63 kg/m^3
Average kinetic flame temperature 1,506.91 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 4.56e+07 W/m^2
Kinetic flame height 4.33 µm
Average kinetic flame density 2.43 kg/m^3
Average kinetic flame temperature 1,635.91 K
Heat flux density due to metal combustion, 2.89e+06 W/m^2
Effective thermal conductivity 0.22 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 1.24e-06 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 5e-05 m
Pore diameter 8.1e-09 m
Diffusion flame heat flux density 1.04e+07 W/m^2
Diffusion flame height 29.21 µm
Heat flux density to surface within skeletal structure 1.52e+07 W/m^2
Heat flux density to surface outside skeletal structure 572,141.31 W/m^2
Total heat flux density to surface 2.61e+07 W/m^2
Binder sublimation heat flux density 2.61e+07 W/m^2
Surface heat balance error -0.0023 W/m^2

<u>*At pressure 4.06 MPa*</u>
Experimental burning rate 37.34 mm/s
Calculated burning rate 28.28 mm/s
First Conditional Fuel
Surface temperature 721.57 K
Linear burning rate 68.8 mm/s
Mass decomposition rate 114.14 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.69e+08 W/m^2
Kinetic flame height 1.76 µm
Average kinetic flame density 7.27 kg/m^3
Average kinetic flame temperature 2,214.28 K
Binder sublimation heat flux density 1.69e+08 W/m^2
Surface heat balance error 0.024 W/m^2
Secondary Conditional Fuel
Surface temperature 655.99 K

Linear burning rate 13.92 mm/s
Mass decomposition rate 23.09 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 5.34e+06 W/m^2
Kinetic flame height 32.03 μm
Average kinetic flame density 10.66 kg/m^3
Average kinetic flame temperature 1,510.5 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 7.96e+07 W/m^2
Kinetic flame height 2.47 μm
Average kinetic flame density 9.82 kg/m^3
Average kinetic flame temperature 1,639.5 K
Heat flux density due to metal combustion, 3.47e+06 W/m^2
Effective thermal conductivity 0.22 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 8.66e-07 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 4.12e-05 m
Pore diameter 5.49e-09 m
Diffusion flame heat flux density 8.51e+06 W/m^2
Diffusion flame height 35.48 μm
Heat flux density to surface within skeletal structure 1.94e+07 W/m^2
Heat flux density to surface outside skeletal structure 4.09e+06 W/m^2
Total heat flux density to surface 3.2e+07 W/m^2
Binder sublimation heat flux density 3.2e+07 W/m^2
Surface heat balance error 0.0026 W/m^2

At pressure 6.5 MPa
Experimental burning rate 50.5 mm/s
Calculated burning rate 30.77 mm/s
First Conditional Fuel
Surface temperature 750 K
Linear burning rate 126.12 mm/s
Mass decomposition rate 209.23 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 3.19e+08 W/m^2
Kinetic flame height 0.93 μm
Average kinetic flame density 11.58 kg/m^3
Average kinetic flame temperature 2,228.5 K
Binder sublimation heat flux density 3.19e+08 W/m^2
Surface heat balance error 0.053 W/m^2
Secondary Conditional Fuel
Surface temperature 656.9 K
Linear burning rate 14.26 mm/s
Mass decomposition rate 23.66 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 1.04e+07 W/m^2
Kinetic flame height 16.44 μm
Average kinetic flame density 17.07 kg/m^3
Average kinetic flame temperature 1,510.95 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1e+08 W/m^2
Kinetic flame height 1.97 μm
Average kinetic flame density 15.73 kg/m^3
Average kinetic flame temperature 1,639.95 K
Heat flux density due to metal combustion, 3.55e+06 W/m^2
Effective thermal conductivity 0.22 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 8.29e-07 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08

Skeleton layer thickness 4.02e-05 m
Pore diameter 5.23e-09 m
Diffusion flame heat flux density 8.3e+06 W/m²
Diffusion flame height 36.35 μm
Heat flux density to surface within skeletal structure 1.57e+07 W/m²
Heat flux density to surface outside skeletal structure 8.82e+06 W/m²
Total heat flux density to surface 3.28e+07 W/m²
Binder sublimation heat flux density 3.28e+07 W/m²
Surface heat balance error 0.002 W/m²

Propellant Composition "Bas_4"

At pressure 1 MPa
Experimental burning rate 9.73 mm/s
Calculated burning rate 10.34 mm/s
First Conditional Fuel
Surface temperature 648.81 K
Linear burning rate 11.46 mm/s
Mass decomposition rate 19.01 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 2.61e+07 W/m²
Kinetic flame height 11.71 μm
Average kinetic flame density 1.82 kg/m³
Average kinetic flame temperature 2,177.91 K
Binder sublimation heat flux density 2.61e+07 W/m²
Surface heat balance error -0.005 W/m²
Secondary Conditional Fuel
Surface temperature 608.78 K
Linear burning rate 3.56 mm/s
Mass decomposition rate 5.9 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 60,559.97 W/m²
Kinetic flame height 715.36 μm
Average kinetic flame density 4.81 kg/m³
Average kinetic flame temperature 825.39 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 3.78e+06 W/m²
Kinetic flame height 16.72 μm
Average kinetic flame density 4.29 kg/m³
Average kinetic flame temperature 924.89 K
Heat flux density due to metal combustion, 6.06e+06 W/m²
Effective thermal conductivity 1.41 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 6.09e-06 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 1.61e-04 m
Pore diameter 8.4e-08 m
Diffusion flame heat flux density 2.53e+06 W/m²
Diffusion flame height 117.58 μm
Heat flux density to surface within skeletal structure 5.2e+06 W/m²
Heat flux density to surface outside skeletal structure 28,537.82 W/m²
Total heat flux density to surface 7.76e+06 W/m²
Binder sublimation heat flux density 7.76e+06 W/m²
Surface heat balance error -8.54e-04 W/m²

At pressure 4.06 MPa
Experimental burning rate 25.92 mm/s
Calculated burning rate 14.99 mm/s
First Conditional Fuel
Surface temperature 721.57 K

Linear burning rate 68.8 mm/s
Mass decomposition rate 114.14 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.69e+08 W/m^2
Kinetic flame height 1.76 μm
Average kinetic flame density 7.27 kg/m^3
Average kinetic flame temperature 2,214.28 K
Binder sublimation heat flux density 1.69e+08 W/m^2
Surface heat balance error 0.024 W/m^2

Secondary Conditional Fuel

Surface temperature 615.33 K
Linear burning rate 4.35 mm/s
Mass decomposition rate 7.22 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 387,561.82 W/m^2
Kinetic flame height 110.09 μm
Average kinetic flame density 19.42 kg/m^3
Average kinetic flame temperature 828.66 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 6.6e+06 W/m^2
Kinetic flame height 9.49 μm
Average kinetic flame density 17.34 kg/m^3
Average kinetic flame temperature 928.16 K
Heat flux density due to metal combustion, 7.34e+06 W/m^2
Effective thermal conductivity 1.41 $W/(m \cdot K)$
Conductive thermal conductivity 1.41 $W/(m \cdot K)$
Radiative thermal conductivity 4.2e-06 $W/(m \cdot K)$
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 1.32e-04 m
Pore diameter 5.61e-08 m

Diffusion flame heat flux density 2.06e+06 W/m^2
Diffusion flame height 143.84 μm
Heat flux density to surface within skeletal structure 7.32e+06 W/m^2
Heat flux density to surface outside skeletal structure 184,099.74 W/m^2
Total heat flux density to surface 9.56e+06 W/m^2
Binder sublimation heat flux density 9.56e+06 W/m^2
Surface heat balance error -0.0011 W/m^2

<u>At pressure 6.5 MPa</u>

Experimental burning rate 36.06 mm/s
Calculated burning rate 16.67 mm/s

First Conditional Fuel

Surface temperature 750 K
Linear burning rate 126.12 mm/s
Mass decomposition rate 209.23 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 3.19e+08 W/m^2
Kinetic flame height 0.93 μm
Average kinetic flame density 11.58 kg/m^3
Average kinetic flame temperature 2,228.5 K
Binder sublimation heat flux density 3.19e+08 W/m^2
Surface heat balance error 0.053 W/m^2

Secondary Conditional Fuel

Surface temperature 618.2 K
Linear burning rate 4.75 mm/s
Mass decomposition rate 7.88 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 711,092.79 W/m^2
Kinetic flame height 59.6 μm
Average kinetic flame density 31.08 kg/m^3
Average kinetic flame temperature 830.1 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 7.81e+06 W/m²

Kinetic flame height 7.97 µm

Average kinetic flame density 27.75 kg/m³

Average kinetic flame temperature 929.6 K

Heat flux density due to metal combustion, 7.97e+06 W/m²

Effective thermal conductivity 1.41 W/(m·K)

Conductive thermal conductivity 1.41 W/(m·K)

Radiative thermal conductivity 3.58e-06 W/(m·K)

Conductive thermal conductivity balance error 9.240271880983641E-10

Skeleton layer thickness 1.21e-04 m

Pore diameter 4.71e-08 m

Diffusion flame heat flux density 1.89e+06 W/m²

Diffusion flame height 156.93 µm

Heat flux density to surface within skeletal structure 8.24e+06 W/m²

Heat flux density to surface outside skeletal structure 340,071.48 W/m²

Total heat flux density to surface 1.05e+07 W/m²

Binder sublimation heat flux density 1.05e+07 W/m²

Surface heat balance error 6.27e-04 W/m²

Propellant Composition "Bas_1"

At pressure 1 MPa

Experimental burning rate 14.26 mm/s

Calculated burning rate 18.26 mm/s

First Conditional Fuel

Surface temperature 602.18 K

Linear burning rate 11.15 mm/s

Mass decomposition rate 18.49 $kg \cdot s^{-1} \cdot m^{-2}$

Kinetic flame heat flux density to surface 2.41e+07 W/m²

Kinetic flame height 12.88 µm

Average kinetic flame density 1.84 kg/m³

Average kinetic flame temperature 2,154.59 K

Binder sublimation heat flux density 2.41e+07 W/m²

Surface heat balance error -0.0051 W/m²

Secondary Conditional Fuel

Surface temperature 600 K

Linear burning rate 10.45 mm/s

Mass decomposition rate 17.34 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 900,610.05 W/m²

Kinetic flame height 195.98 µm

Average kinetic flame density 2.68 kg/m³

Average kinetic flame temperature 1,482.5 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 9.09e+06 W/m²

Kinetic flame height 10.91 µm

Average kinetic flame density 3.62 kg/m³

Average kinetic flame temperature 1,095.5 K

Heat flux density due to metal combustion, 4.49e+07 W/m²

Effective thermal conductivity 0.4 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 9.31e-07 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 6.29e-06 m

Pore diameter 1.07e-08 m

Diffusion flame heat flux density 1.58e+06 W/m²

Diffusion flame height 194.18 µm

Heat flux density to surface within skeletal structure 2.04e+07 W/m²

Heat flux density to surface outside skeletal structure 560,130.55 W/m²
Total heat flux density to surface 2.25e+07 W/m²
Binder sublimation heat flux density 2.25e+07 W/m²
Surface heat balance error 8.48e-04 W/m²

At pressure 4.06 MPa
Experimental burning rate 38.01 mm/s
Calculated burning rate 29.84 mm/s
First Conditional Fuel
Surface temperature 669.47 K
Linear burning rate 66.75 mm/s
Mass decomposition rate 110.74 $kg\cdot s^{-1}\cdot m^{-2}$
Kinetic flame heat flux density to surface 1.56e+08 W/m²
Kinetic flame height 1.95 µm
Average kinetic flame density 7.36 kg/m³
Average kinetic flame temperature 2,188.23 K
Binder sublimation heat flux density 1.56e+08 W/m²
Surface heat balance error 0.031 W/m²
Secondary Conditional Fuel
Surface temperature 608 K
Linear burning rate 13.22 mm/s
Mass decomposition rate 21.93 $kg\cdot s^{-1}\cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 5.54e+06 W/m²
Kinetic flame height 31.69 µm
Average kinetic flame density 10.83 kg/m³
Average kinetic flame temperature 1,486.5 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 5.92e+07 W/m²
Kinetic flame height 1.66 µm
Average kinetic flame density 14.64 kg/m³
Average kinetic flame temperature 1,099.5 K
Heat flux density due to metal combustion, 5.61e+07 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 6.05e-07 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 4.97e-06 m
Pore diameter 6.67e-09 m
Diffusion flame heat flux density 1.24e+06 W/m²
Diffusion flame height 245.66 µm
Heat flux density to surface within skeletal structure 2.31e+07 W/m²
Heat flux density to surface outside skeletal structure 4.43e+06 W/m²
Total heat flux density to surface 2.88e+07 W/m²
Binder sublimation heat flux density 2.88e+07 W/m²
Surface heat balance error 2.85e-04 W/m²

At pressure 6.5 MPa
Experimental burning rate 52.88 mm/s
Calculated burning rate 38.01 mm/s
First Conditional Fuel
Surface temperature 695.75 K
Linear burning rate 122.23 mm/s
Mass decomposition rate 202.78 $kg\cdot s^{-1}\cdot m^{-2}$
Kinetic flame heat flux density to surface 2.93e+08 W/m²
Kinetic flame height 1.03 µm
Average kinetic flame density 11.72 kg/m³
Average kinetic flame temperature 2,201.38 K
Binder sublimation heat flux density 2.93e+08 W/m²

Surface heat balance error 0.026 W/m²

Secondary Conditional Fuel

Surface temperature 615.56 K

Linear burning rate 16.42 mm/s

Mass decomposition rate 27.23 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 8.93e+06 W/m²

Kinetic flame height 19.6 µm

Average kinetic flame density 17.31 kg/m³

Average kinetic flame temperature 1,490.28 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1e+08 W/m²

Kinetic flame height 0.98 µm

Average kinetic flame density 23.38 kg/m³

Average kinetic flame temperature 1,103.28 K

Heat flux density due to metal combustion, 6.9e+07 W/m²

Effective thermal conductivity 0.4 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 4.07e-07 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 4.01e-06 m

Pore diameter 4.33e-09 m

Diffusion flame heat flux density 1e+06 W/m²

Diffusion flame height 305.04 µm

Heat flux density to surface within skeletal structure 2.76e+07 W/m²

Heat flux density to surface outside skeletal structure 7.47e+06 W/m²

Total heat flux density to surface 3.61e+07 W/m²

Binder sublimation heat flux density 3.61e+07 W/m²

Surface heat balance error -9.28e-04 W/m²

Propellant Composition "Bas_0"

<u>*At pressure 1 MPa*</u>

Experimental burning rate 5.97 mm/s

Calculated burning rate 7.41 mm/s

First Conditional Fuel

Surface temperature 648.81 K

Linear burning rate 11.46 mm/s

Mass decomposition rate 19.01 kg·s⁻¹·m⁻²

Kinetic flame heat flux density to surface 2.61e+07 W/m²

Kinetic flame height 11.71 µm

Average kinetic flame density 1.82 kg/m³

Average kinetic flame temperature 2,177.91 K

Binder sublimation heat flux density 2.61e+07 W/m²

Surface heat balance error 0.003 W/m²

Secondary Conditional Fuel

Surface temperature 607.42 K

Linear burning rate 3.41 mm/s

Mass decomposition rate 5.66 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 2.77e+06 W/m²

Kinetic flame height 63.56 µm

Average kinetic flame density 2.67 kg/m³

Average kinetic flame temperature 1,486.21 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 48,251.52 W/m²

Kinetic flame height 2,038.45 µm

Average kinetic flame density 3.61 kg/m³

Average kinetic flame temperature 1,099.21 K

Heat flux density due to metal combustion, 2.07e+06 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 7.78e-06 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.35e-04 m
Pore diameter 8.6e-08 m
Diffusion flame heat flux density 4.83e+06 W/m²
Diffusion flame height 63.39 µm
Heat flux density to surface within skeletal structure 547,912.62 W/m²
Heat flux density to surface outside skeletal structure 2.05e+06 W/m²
Total heat flux density to surface 7.42e+06 W/m²
Binder sublimation heat flux density 7.42e+06 W/m²
Surface heat balance error 0.0017 W/m²

At pressure 4.06 MPa
Experimental burning rate 13.63 mm/s
Calculated burning rate 14.19 mm/s
First Conditional Fuel
Surface temperature 721.57 K
Linear burning rate 68.8 mm/s
Mass decomposition rate 114.14 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.69e+08 W/m²
Kinetic flame height 1.76 µm
Average kinetic flame density 7.27 kg/m³
Average kinetic flame temperature 2,214.28 K
Binder sublimation heat flux density 1.69e+08 W/m²
Surface heat balance error -0.0096 W/m²
Secondary Conditional Fuel
Surface temperature 626.11 K
Linear burning rate 6.01 mm/s
Mass decomposition rate 9.98 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 1.23e+07 W/m²
Kinetic flame height 14.19 µm
Average kinetic flame density 10.76 kg/m³
Average kinetic flame temperature 1,495.55 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 137,231.39 W/m²
Kinetic flame height 703.11 µm
Average kinetic flame density 14.52 kg/m³
Average kinetic flame temperature 1,108.55 K
Heat flux density due to metal combustion, 3.55e+06 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 2.74e-06 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 7.66e-05 m
Pore diameter 2.77e-08 m
Diffusion flame heat flux density 2.72e+06 W/m²
Diffusion flame height 111.75 µm
Heat flux density to surface within skeletal structure 691,886.01 W/m²
Heat flux density to surface outside skeletal structure 9.95e+06 W/m²
Total heat flux density to surface 1.34e+07 W/m²
Binder sublimation heat flux density 1.34e+07 W/m²
Surface heat balance error 0.0019 W/m²

At pressure 6.5 MPa
Experimental burning rate 18 mm/s

Calculated burning rate 19.47 mm/s

First Conditional Fuel

- Surface temperature 750 K
- Linear burning rate 126.12 mm/s
- Mass decomposition rate 209.23 $kg \cdot s^{-1} \cdot m^{-2}$
- Kinetic flame heat flux density to surface 3.19e+08 W/m^2
- Kinetic flame height 0.93 μm
- Average kinetic flame density 11.58 kg/m^3
- Average kinetic flame temperature 2,228.5 K
- Binder sublimation heat flux density 3.19e+08 W/m^2
- Surface heat balance error 0.011 W/m^2

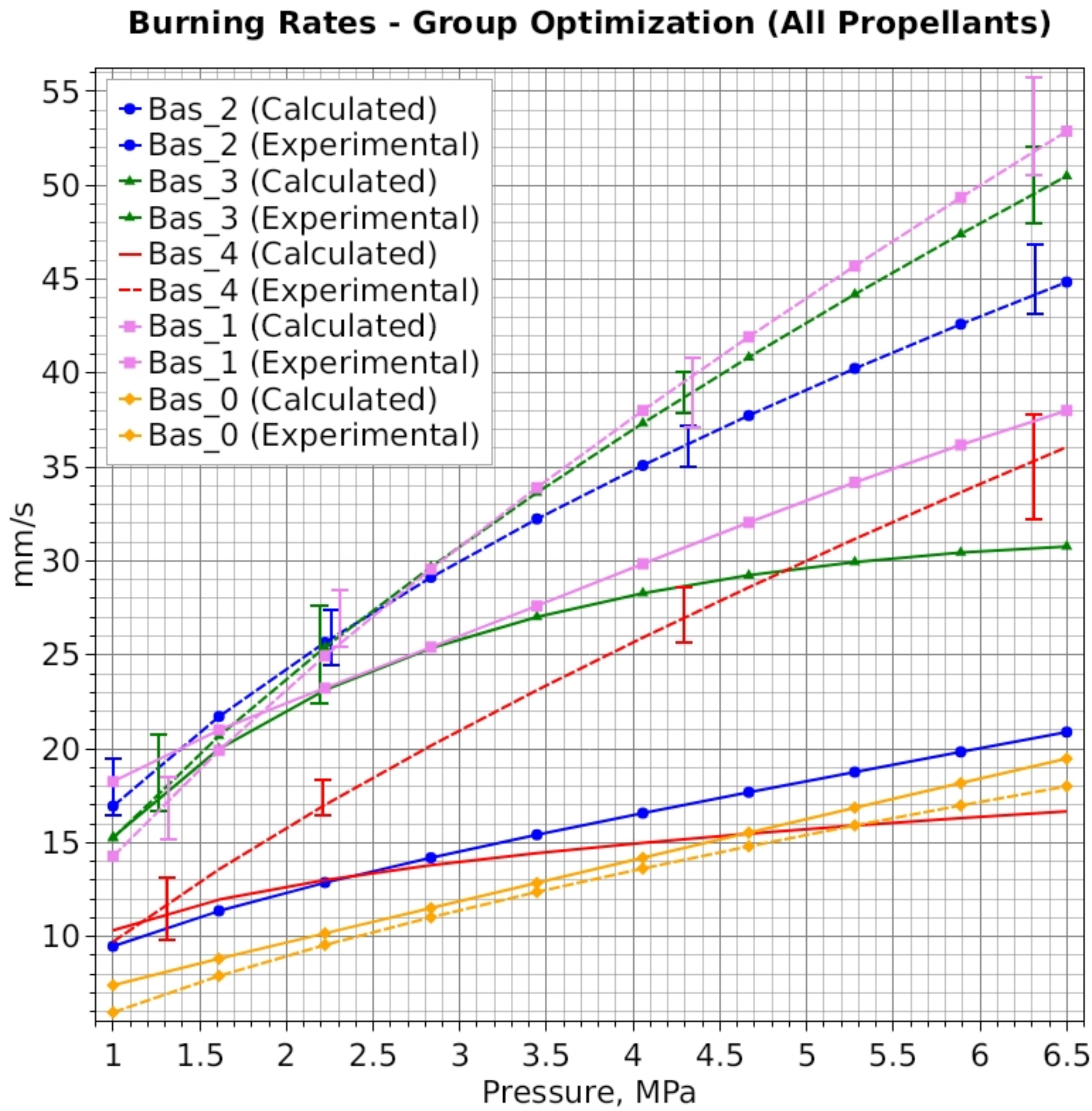
Secondary Conditional Fuel

- Surface temperature 636.71 K
- Linear burning rate 8.17 mm/s
- Mass decomposition rate 13.56 $kg \cdot s^{-1} \cdot m^{-2}$
- *Region Outside Skeleton Structure*
  - Kinetic flame heat flux density to surface 1.8e+07 W/m^2
  - Kinetic flame height 9.58 μm
  - Average kinetic flame density 17.19 kg/m^3
  - Average kinetic flame temperature 1,500.86 K
- *Skeleton Structure Region*
  - Kinetic flame heat flux density to surface 180,399.09 W/m^2
  - Kinetic flame height 528.99 μm
  - Average kinetic flame density 23.16 kg/m^3
  - Average kinetic flame temperature 1,113.86 K
  - Heat flux density due to metal combustion, 4.75e+06 W/m^2
  - Effective thermal conductivity 0.4 W/(m·K)
  - Conductive thermal conductivity 0.4 W/(m·K)
  - Radiative thermal conductivity 1.56e-06 W/(m·K)
  - Conductive thermal conductivity balance error -4.092294297874943E-08
  - Skeleton layer thickness 5.63e-05 m
  - Pore diameter 1.5e-08 m
- Diffusion flame heat flux density 1.99e+06 W/m^2
- Diffusion flame height 151.87 μm
- Heat flux density to surface within skeletal structure 621,750.52 W/m^2
- Heat flux density to surface outside skeletal structure 1.58e+07 W/m^2
- Total heat flux density to surface 1.84e+07 W/m^2
- Binder sublimation heat flux density 1.84e+07 W/m^2
- Surface heat balance error 9.73e-05 W/m^2

1 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	16.94 mm/s		15.24 mm/s		9.73 mm/s		14.26 mm/s		5.97 mm/s	
Calculated Burning Rate	9.49 mm/s		15.3 mm/s		10.34 mm/s		18.26 mm/s		7.41 mm/s	
Difference	7.46 mm/s		-0.055 mm/s		-0.61 mm/s		-3.99 mm/s		-1.44 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	11.46 mm/s	4.53 mm/s	11.46 mm/s	11.46 mm/s	11.46 mm/s	3.56 mm/s	11.15 mm/s	10.45 mm/s	11.46 mm/s	3.41 mm/s
Surface Temperature	648.81 K	616.62 K	648.81 K	648.81 K	648.81 K	608.78 K	602.18 K	600 K	648.81 K	607.42 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	2.61e+07 W/m²		2.61e+07 W/m²		2.61e+07 W/m²		2.41e+07 W/m²		2.61e+07 W/m²	
Kinetic Flame Heat Flux Density (Skeleton)	1.2e+07 W/m²		4.56e+07 W/m²		3.78e+06 W/m²		9.09e+06 W/m²		48,251.52 W/m²	
Kinetic Flame Height (Skeleton)	8.15 μm		4.33 μm		16.72 μm		10.91 μm		2,038.45 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	2.09e+06 W/m²		832,702.72 W/m²		60,559.97 W/m²		900,610.05 W/m²		2.77e+06 W/m²	
Kinetic Flame Height (Out-Skeleton)	83.73 μm		206.1 μm		715.36 μm		195.98 μm		63.56 μm	
Diffusion Flame Heat Flux Density	3.55e+06 W/m²		1.04e+07 W/m²		2.53e+06 W/m²		1.58e+06 W/m²		4.83e+06 W/m²	
Diffusion Flame Height	84.17 μm		29.21 μm		117.58 μm		194.18 μm		63.39 μm	
Metal Combustion Heat Flux Density	2.18e+06 W/m²		2.89e+06 W/m²		6.06e+06 W/m²		4.49e+07 W/m²		2.07e+06 W/m²	

4.06 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	35.09 mm/s		37.34 mm/s		25.92 mm/s		38.01 mm/s		13.63 mm/s	
Calculated Burning Rate	16.57 mm/s		28.28 mm/s		14.99 mm/s		29.84 mm/s		14.19 mm/s	
Difference	18.52 mm/s		9.06 mm/s		10.93 mm/s		8.17 mm/s		-0.57 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	68.8 mm/s	7.06 mm/s	68.8 mm/s	13.92 mm/s	68.8 mm/s	4.35 mm/s	66.75 mm/s	13.22 mm/s	68.8 mm/s	6.01 mm/s
Surface Temperature	721.57 K	631.63 K	721.57 K	655.99 K	721.57 K	615.33 K	669.47 K	608 K	721.57 K	626.11 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.69e+08 W/m²		1.69e+08 W/m²		1.69e+08 W/m²		1.56e+08 W/m²		1.69e+08 W/m²	
Kinetic Flame Heat Flux Density (Skeleton)	1.65e+07 W/m²		7.96e+07 W/m²		6.6e+06 W/m²		5.92e+07 W/m²		137,231.39 W/m²	
Kinetic Flame Height (Skeleton)	5.82 µm		2.47 µm		9.49 µm		1.66 µm		703.11 µm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	1.04e+07 W/m²		5.34e+06 W/m²		387,561.82 W/m²		5.54e+06 W/m²		1.23e+07 W/m²	
Kinetic Flame Height (Out-Skeleton)	16.6 µm		32.03 µm		110.09 µm		31.69 µm		14.19 µm	
Diffusion Flame Heat Flux Density	2.26e+06 W/m²		8.51e+06 W/m²		2.06e+06 W/m²		1.24e+06 W/m²		2.72e+06 W/m²	
Diffusion Flame Height	131.28 µm		35.48 µm		143.84 µm		245.66 µm		111.75 µm	
Metal Combustion Heat Flux Density	3.33e+06 W/m²		3.47e+06 W/m²		7.34e+06 W/m²		5.61e+07 W/m²		3.55e+06 W/m²	

6.5 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	44.85 mm/s		50.5 mm/s		36.06 mm/s		52.88 mm/s		18 mm/s	
Calculated Burning Rate	20.9 mm/s		30.77 mm/s		16.67 mm/s		38.01 mm/s		19.47 mm/s	
Difference	23.95 mm/s		19.72 mm/s		19.39 mm/s		14.87 mm/s		-1.47 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	126.12 mm/s	8.79 mm/s	126.12 mm/s	14.26 mm/s	126.12 mm/s	4.75 mm/s	122.23 mm/s	16.42 mm/s	126.12 mm/s	8.17 mm/s
Surface Temperature	750 K	639.27 K	750 K	656.9 K	750 K	618.2 K	695.75 K	615.56 K	750 K	636.71 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	3.19e+08 W/m²		3.19e+08 W/m²		3.19e+08 W/m²		2.93e+08 W/m²		3.19e+08 W/m²	
Kinetic Flame Heat Flux Density (Skeleton)	1.72e+07 W/m²		1e+08 W/m²		7.81e+06 W/m²		1e+08 W/m²		180,399.09 W/m²	
Kinetic Flame Height (Skeleton)	5.54 µm		1.97 µm		7.97 µm		0.98 µm		528.99 µm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	1.68e+07 W/m²		1.04e+07 W/m²		711,092.79 W/m²		8.93e+06 W/m²		1.8e+07 W/m²	
Kinetic Flame Height (Out-Skeleton)	10.28 µm		16.44 µm		59.6 µm		19.6 µm		9.58 µm	
Diffusion Flame Heat Flux Density	1.82e+06 W/m²		8.3e+06 W/m²		1.89e+06 W/m²		1e+06 W/m²		1.99e+06 W/m²	
Diffusion Flame Height	163.31 µm		36.35 µm		156.93 µm		305.04 µm		151.87 µm	
Metal Combustion Heat Flux Density	4.09e+06 W/m²		3.55e+06 W/m²		7.97e+06 W/m²		6.9e+07 W/m²		4.75e+06 W/m²	



Propellant Composition Parameters

Composition "Bas_2"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: U = A · p^v, A = 1.285341E-05; v = 0.52.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,604 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_3"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.57.

Vieille's Power Law: U = A · p^v, A = 2.203002E-06; v = 0.64.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,674 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,623 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_4"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.56.

Vieille's Power Law: U = A · p^v, A = 6.137894E-07; v = 0.7.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,580 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,241 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,042 K

Composition "Bas_1"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: U = A · p^v, A = 9.000448E-07; v = 0.7.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_0"

Density 1,659 kg/m^3, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.
Vieille's Power Law: U = A · p^v, A = 1.720582E-06; v = 0.59.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

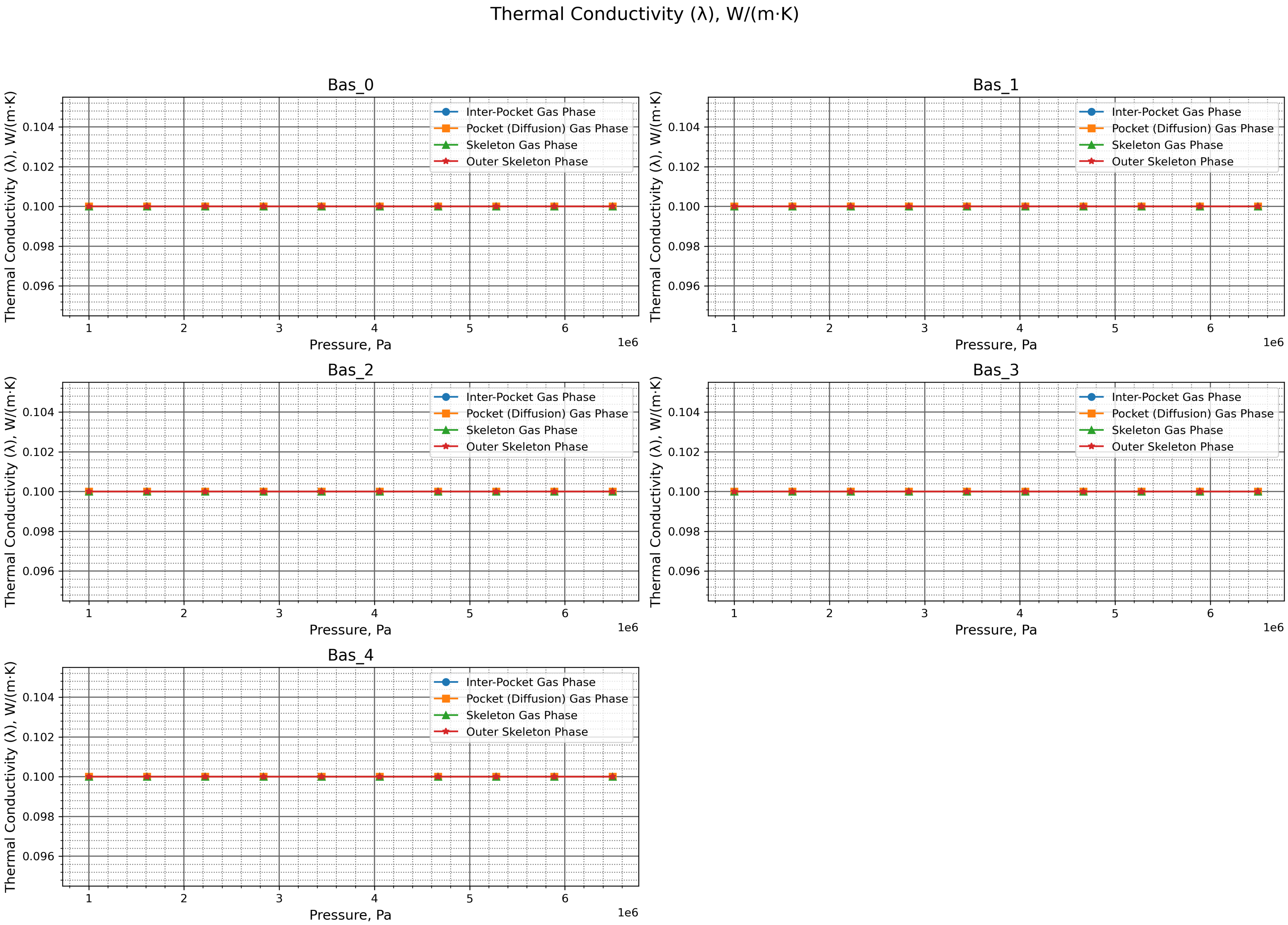
Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,666 K

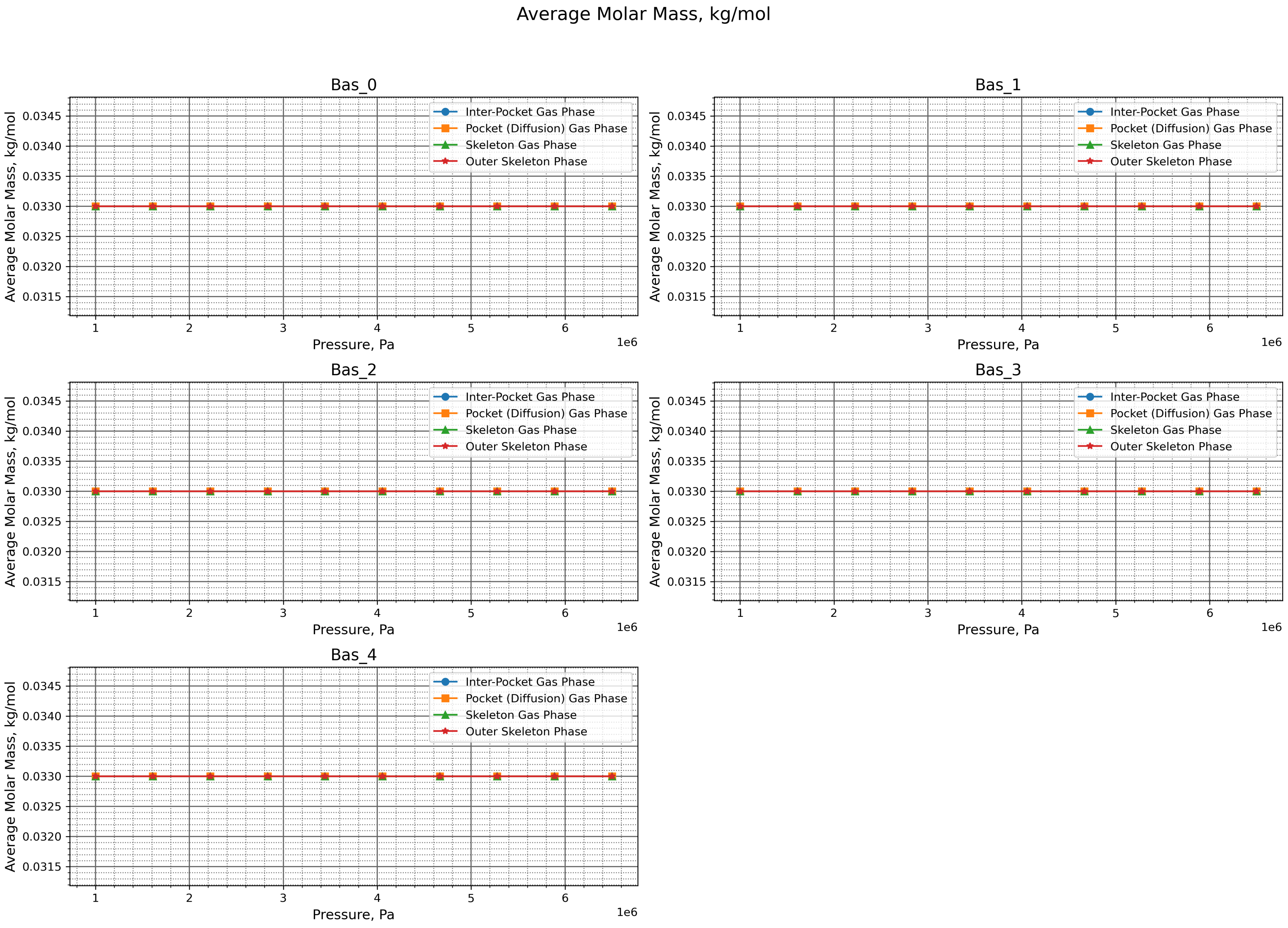
Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,591 K

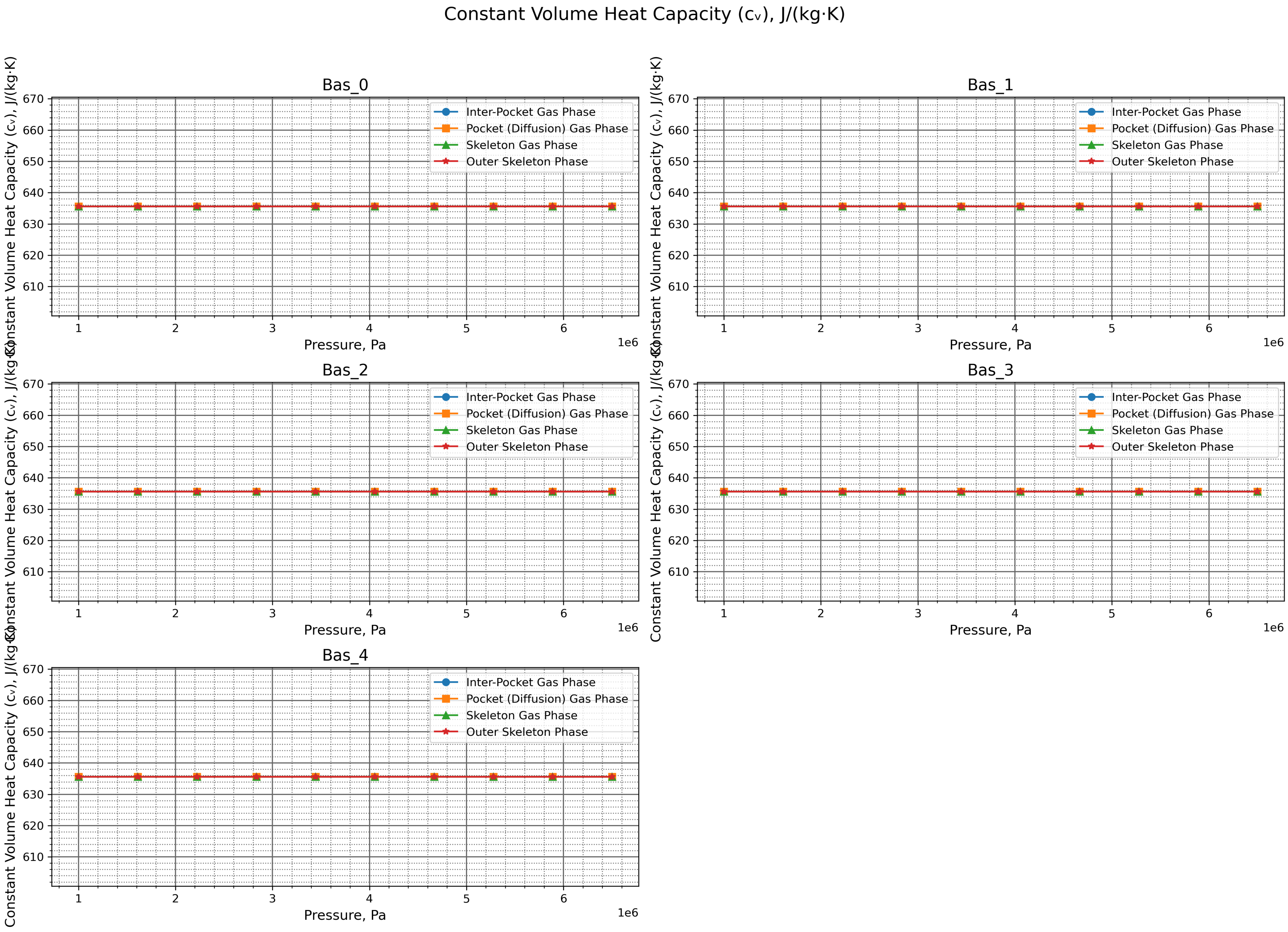
Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

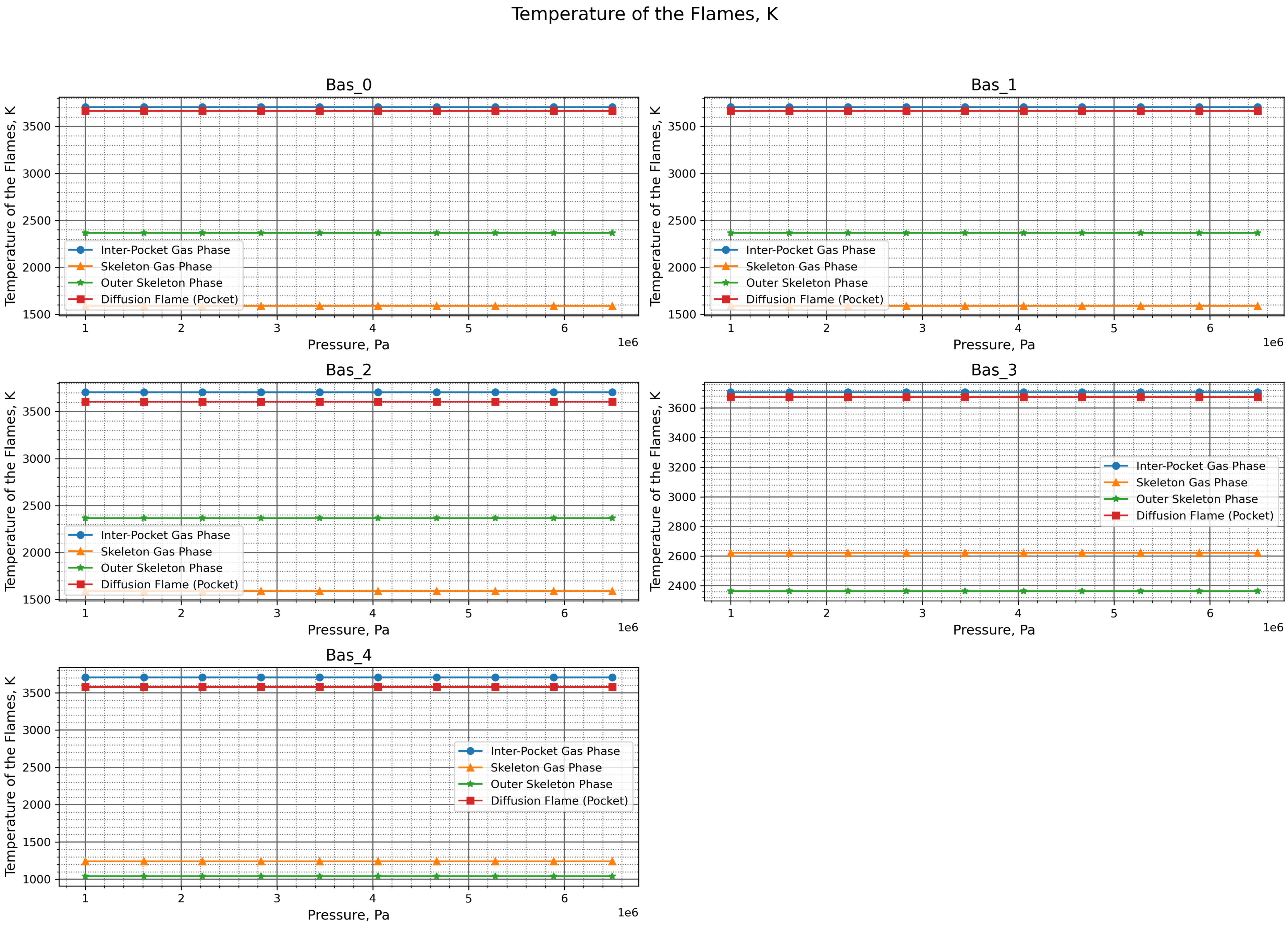

Thermal Conductivity (λ), W/(m·K)
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4
Inter-Pocket Gas Phase
Pocket (Diffusion) Gas Phase
Skeleton Gas Phase
Outer Skeleton Phase
Thermal Conductivity (λ), W/(m·K)
Pressure, Pa
1e6
0.104
0.102
0.100
0.098
0.096
1
2
3
4
5
6



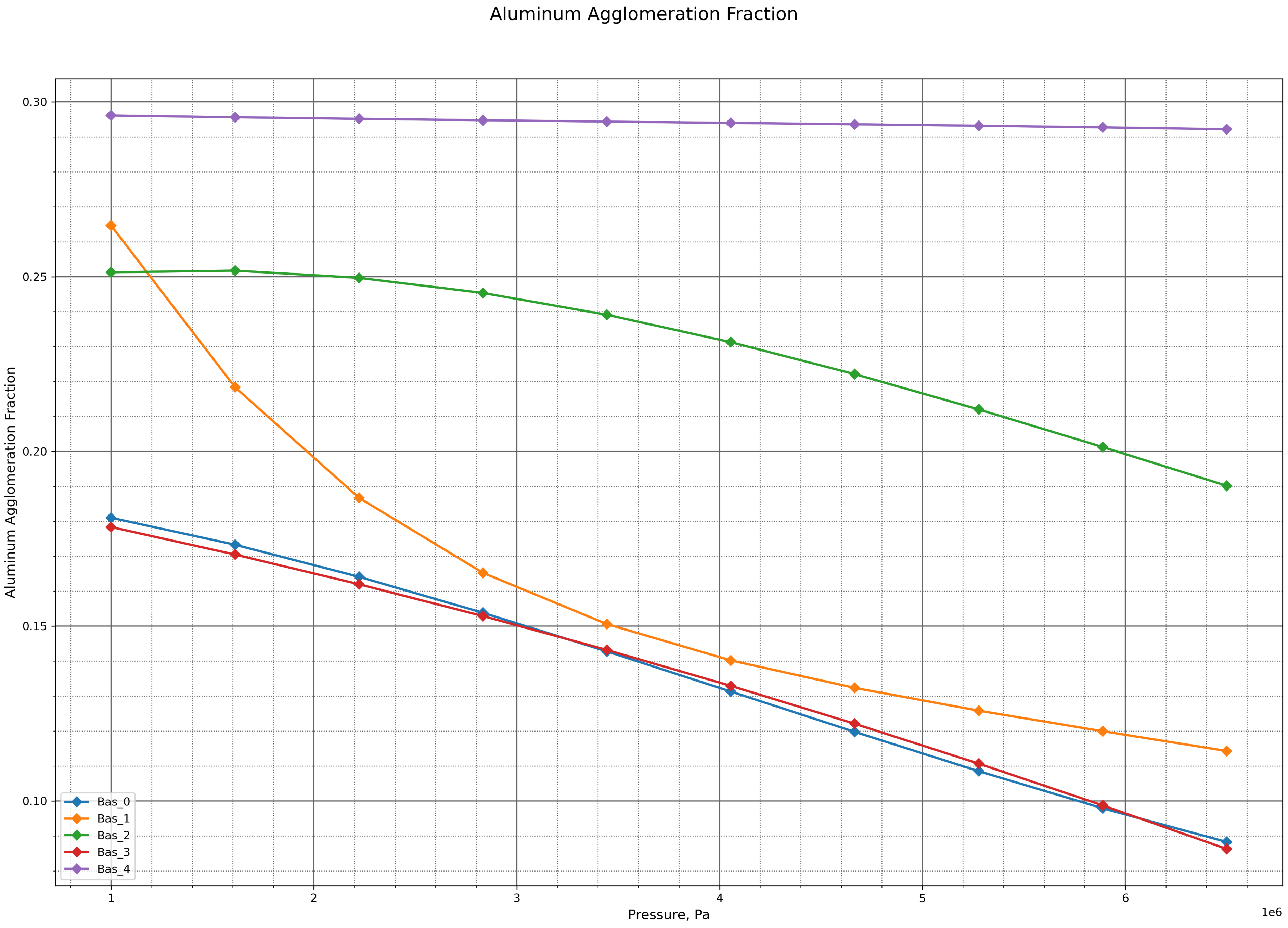
Average Molar Mass, kg/mol
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4
Inter-Pocket Gas Phase
Pocket (Diffusion) Gas Phase
Skeleton Gas Phase
Outer Skeleton Phase
Average Molar Mass, kg/mol
Pressure, Pa
1e6
0.0315
0.0320
0.0325
0.0330
0.0335
0.0340
0.0345
1
2
3
4
5
6



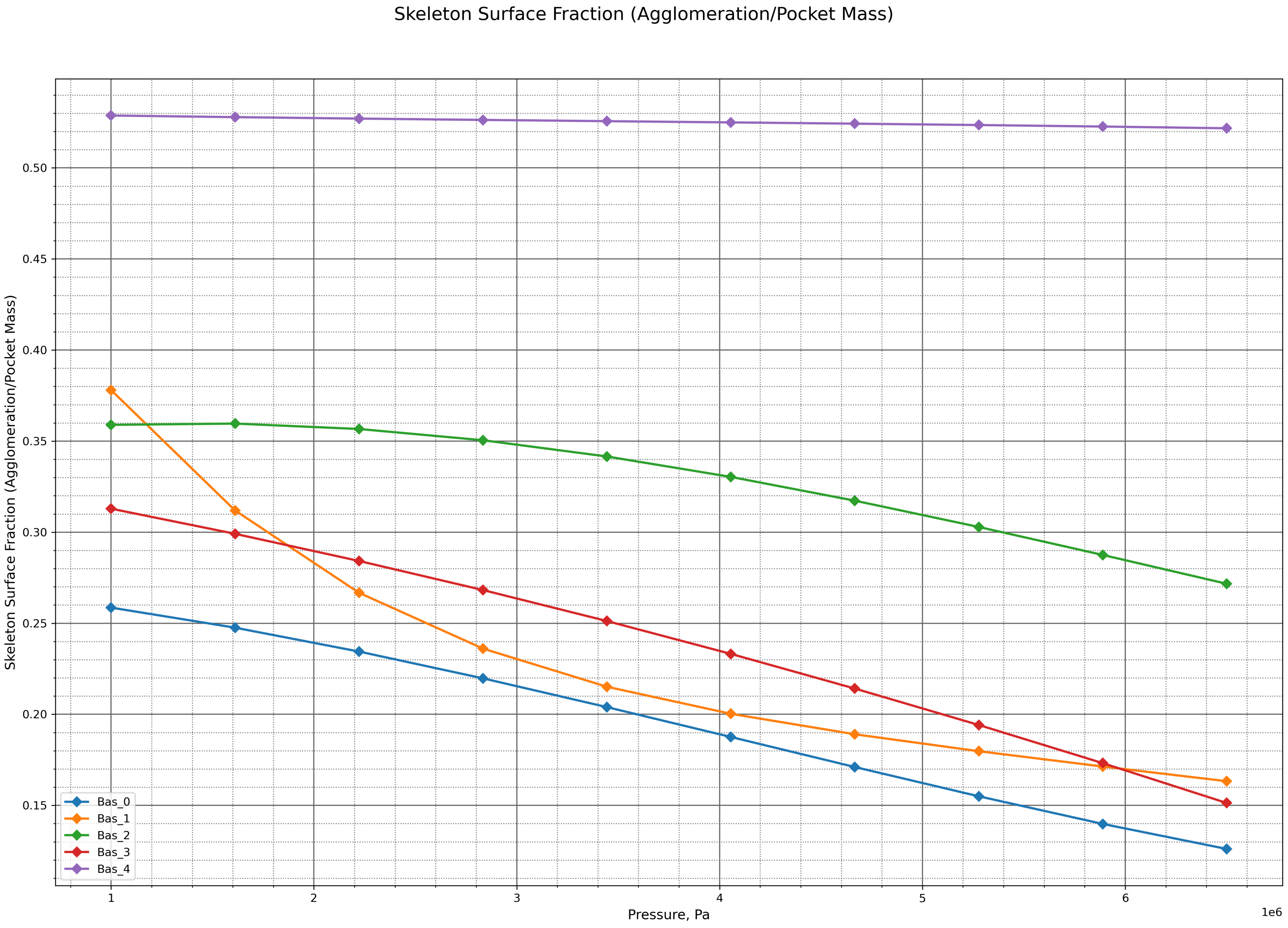
Constant Volume Heat Capacity (cᵥ), J/(kg·K)
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4
Inter-Pocket Gas Phase
Pocket (Diffusion) Gas Phase
Skeleton Gas Phase
Outer Skeleton Phase
Constant Volume Heat Capacity (cᵥ), J/(kg·K)
Pressure, Pa
1e6
670
660
650
640
630
620
610
1
2
3
4
5
6



Temperature of the Flames, K
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4
Inter-Pocket Gas Phase
Skeleton Gas Phase
Outer Skeleton Phase
Diffusion Flame (Pocket)
Temperature of the Flames, K
Pressure, Pa
1e6



Aluminum Agglomeration Fraction
Aluminum Agglomeration Fraction
0.30
0.25
0.20
0.15
0.10
1
2
3
4
5
6
1e6
Pressure, Pa
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4



Skeleton Surface Fraction (Agglomeration/Pocket Mass)
Skeleton Surface Fraction (Agglomeration/Pocket Mass)
Pressure, Pa
1e6
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15
1
2
3
4
5
6
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4

Performance Report

System Information
Operating System: Unix 6.8.0.117 (Ubuntu 24.04.4 LTS)
Processor: Intel(R) Xeon(R) CPU E5-2697 v4 @ 2.30GHz (Base Clock Frequency 2.29 GHz)
Memory: 15.6 GB (64-bit), Unknown
Machine Name: vscode
Runtime: .NET 10.0.8 (.NET 10.0.8)
Architecture: X64 / X64

Total Execution Time
Total execution time: 00:27:42.413

Garbage Collection Information
Total Memory: 8938144 bytes
GC Pause Time Percentage: 0.13%
Generation 0 Collections: 1956
Generation 1 Collections: 166
Generation 2 Collections: 17

Detailed Performance Metrics
Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance
Call count: 1442196
Maximum execution time: 17.1747 ms
Minimum execution time: 0.0001 ms
Mean execution time: 0.6135839207014864 ms
Execution time standard deviation: 0.34603441127373985 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance
Call count: 912741
Maximum execution time: 6.3441 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18838952167153789 ms
Execution time standard deviation: 0.13451698378324298 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance
Call count: 592693
Maximum execution time: 3.7864 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2658289087267809 ms
Execution time standard deviation: 0.06564296272309951 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 8.7026 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.614030496139567 ms
Execution time standard deviation: 0.3444042238607047 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance
Call count: 913540
Maximum execution time: 7.3961 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1882743947719881 ms
Execution time standard deviation: 0.13479098381873825 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance
Call count: 592184
Maximum execution time: 7.4649 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.2657417336503482 ms
Execution time standard deviation: 0.06569053102626 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 12.5691 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6225389430799516 ms
Execution time standard deviation: 0.3466582665206925 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 914160
Maximum execution time: 5.6138 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.19141305548262819 ms
Execution time standard deviation: 0.13546202750868444 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 593695
Maximum execution time: 3.9899 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.269642510548343 ms
Execution time standard deviation: 0.06239201475963392 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 8.2363 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6167052909117193 ms
Execution time standard deviation: 0.34441347743418776 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 913217
Maximum execution time: 4.5652 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1881978243944264 ms
Execution time standard deviation: 0.1340195355488328 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 591799
Maximum execution time: 5.5567 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.26664678919700013 ms
Execution time standard deviation: 0.06381444859600295 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 8.5674 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6180416495354435 ms
Execution time standard deviation: 0.34489017865092736 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 913477
Maximum execution time: 4.1399 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18939081181025874 ms
Execution time standard deviation: 0.1340705452073311 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 593056
Maximum execution time: 3.9653 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.26756434080424163 ms
Execution time standard deviation: 0.0631121911545021 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 11.6424 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6097694827758594 ms
Execution time standard deviation: 0.3434849379762871 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance
Call count: 913557
Maximum execution time: 4.3831 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18701872844278553 ms
Execution time standard deviation: 0.13368961233068852 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance
Call count: 592324
Maximum execution time: 3.5234 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.26406149573543847 ms
Execution time standard deviation: 0.0649012492240195 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 12.8588 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6300680202412258 ms
Execution time standard deviation: 0.3486216715012003 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance
Call count: 914124
Maximum execution time: 3.9477 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1928891220447011 ms
Execution time standard deviation: 0.13586861809345754 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance
Call count: 593082
Maximum execution time: 6.8313 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2721924007135581 ms
Execution time standard deviation: 0.06208199196233107 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 8.338 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6157547799574535 ms
Execution time standard deviation: 0.34416373134989675 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance
Call count: 914136

Maximum execution time: 3.9096 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18857362394654367 ms
Execution time standard deviation: 0.13381381927025288 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance
Call count: 593037
Maximum execution time: 4.9535 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2662568755406638 ms
Execution time standard deviation: 0.06466349183642416 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 26.562 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6096348152879961 ms
Execution time standard deviation: 0.3441566351618173 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance
Call count: 912175
Maximum execution time: 4.3372 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1869268598678996 ms
Execution time standard deviation: 0.13307355347428404 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance
Call count: 592412
Maximum execution time: 4.3039 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.26389491502535845 ms
Execution time standard deviation: 0.06382926870891956 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 25.6476 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6103583385351041 ms
Execution time standard deviation: 0.344884897753459 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance
Call count: 914496
Maximum execution time: 13.4567 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18755506650657988 ms
Execution time standard deviation: 0.13537886495819768 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance
Call count: 593878
Maximum execution time: 9.3236 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2644576520766948 ms
Execution time standard deviation: 0.06708515758234602 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 17.2932 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6065788893420405 ms

Execution time standard deviation: 0.342622280950967 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance
Call count: 913478
Maximum execution time: 3.3973 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1864509882011387 ms
Execution time standard deviation: 0.1327332955303483 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance
Call count: 593356
Maximum execution time: 17.8181 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.26362856463911066 ms
Execution time standard deviation: 0.06903368458915801 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 14.5645 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6130690300122277 ms
Execution time standard deviation: 0.3440666459360244 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance
Call count: 914172
Maximum execution time: 6.3669 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1878305213898398 ms
Execution time standard deviation: 0.13357448173277922 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance
Call count: 593481
Maximum execution time: 8.895 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.26569898075927784 ms
Execution time standard deviation: 0.06511597080216056 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 16.2742 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6146828607335433 ms
Execution time standard deviation: 0.34437311678286614 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance
Call count: 914803
Maximum execution time: 5.1904 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18867171008403194 ms
Execution time standard deviation: 0.1336592692935884 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance
Call count: 594427
Maximum execution time: 7.0298 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.265725761784037 ms
Execution time standard deviation: 0.06397995012435195 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 1441812
Maximum execution time: 15.4194 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6155108051535303 ms
Execution time standard deviation: 0.34607023425777284 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance
Call count: 914249
Maximum execution time: 10.3209 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18935350096090847 ms
Execution time standard deviation: 0.13578154093191105 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance
Call count: 593860
Maximum execution time: 5.2471 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2660044468393207 ms
Execution time standard deviation: 0.06561998711841587 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 16.8897 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6096112130430389 ms
Execution time standard deviation: 0.3424909959812431 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance
Call count: 914295
Maximum execution time: 4.9904 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18683010264738556 ms
Execution time standard deviation: 0.1327680745205373 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance
Call count: 593346
Maximum execution time: 11.1892 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.26391696008736854 ms
Execution time standard deviation: 0.0648944793549303 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 15.672 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6186692263623955 ms
Execution time standard deviation: 0.3455680794961362 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance
Call count: 913570
Maximum execution time: 3.878 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18920280317873897 ms
Execution time standard deviation: 0.13393837957231244 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance
Call count: 593412
Maximum execution time: 11.3748 ms
Minimum execution time: 0.0002 ms

Mean execution time: 0.2671333907976226 ms
Execution time standard deviation: 0.06597631743800268 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 9.966 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6177005565912814 ms
Execution time standard deviation: 0.3442375333456154 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 913117
Maximum execution time: 4.0096 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18933657844503346 ms
Execution time standard deviation: 0.13396021577745576 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 592886
Maximum execution time: 3.1593 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2670647104164942 ms
Execution time standard deviation: 0.06210641472689971 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 15.176 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6173160237257184 ms
Execution time standard deviation: 0.3456911344730263 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 912730
Maximum execution time: 15.4415 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18933289789970803 ms
Execution time standard deviation: 0.13592077980933973 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 591867
Maximum execution time: 4.1251 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.26744373144641326 ms
Execution time standard deviation: 0.06284143019044969 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 14.3475 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6075519864586538 ms
Execution time standard deviation: 0.34329364179952876 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 911960
Maximum execution time: 10.2604 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.186586454669062 ms
Execution time standard deviation: 0.1335412971473326 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 591966
Maximum execution time: 8.3354 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2634879988039898 ms
Execution time standard deviation: 0.06560120792366969 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 9.5194 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6010579484704021 ms
Execution time standard deviation: 0.34097033016679157 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 913430
Maximum execution time: 5.6613 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1850978362874015 ms
Execution time standard deviation: 0.13260316849037987 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 593380
Maximum execution time: 3.6118 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.26067323940813103 ms
Execution time standard deviation: 0.06522547865880261 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 32.8853 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6220249330702946 ms
Execution time standard deviation: 0.3475837648273246 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 913868
Maximum execution time: 6.9473 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1905099425737522 ms
Execution time standard deviation: 0.13505039515430184 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 593307
Maximum execution time: 14.3334 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.27018367994984327 ms
Execution time standard deviation: 0.06694059947946772 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 10.0948 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.5996965857546158 ms
Execution time standard deviation: 0.34106200732753755 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 914257
Maximum execution time: 3.9617 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.18398316862764127 ms
Execution time standard deviation: 0.13175479965305448 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 593841
Maximum execution time: 7.325 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.25995284259592655 ms
Execution time standard deviation: 0.0674196967089386 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 9.8104 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6134885387970225 ms
Execution time standard deviation: 0.34323973693166554 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 914835
Maximum execution time: 5.9131 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18874594391338426 ms
Execution time standard deviation: 0.13404635320872466 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 593870
Maximum execution time: 3.5042 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.26537338020105167 ms
Execution time standard deviation: 0.0634431754100737 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 14.7847 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.615467115962403 ms
Execution time standard deviation: 0.3443800115297419 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 913026
Maximum execution time: 5.1997 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18893851544205073 ms
Execution time standard deviation: 0.13371012310193006 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 592686
Maximum execution time: 20.3319 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2666049937066167 ms
Execution time standard deviation: 0.06803880949435746 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 11.8181 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6115065967684998 ms
Execution time standard deviation: 0.34332814424967506 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance
Call count: 914601
Maximum execution time: 6.5162 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18711496565169244 ms
Execution time standard deviation: 0.13367873344850673 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance
Call count: 593224
Maximum execution time: 5.2431 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.26440000539424646 ms
Execution time standard deviation: 0.06383999451915894 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 15.2652 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6235527641606926 ms
Execution time standard deviation: 0.3448971043585014 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance
Call count: 914448
Maximum execution time: 7.3347 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.19081109478066827 ms
Execution time standard deviation: 0.1343879767502808 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance
Call count: 593136
Maximum execution time: 7.2005 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.26969070179520216 ms
Execution time standard deviation: 0.06080806585869098 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 9.2101 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6024748404784893 ms
Execution time standard deviation: 0.3415353683605883 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance
Call count: 913433
Maximum execution time: 4.1241 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18497366440669805 ms
Execution time standard deviation: 0.13214877430387273 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance
Call count: 593435
Maximum execution time: 15.3624 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.26080724679198114 ms
Execution time standard deviation: 0.06906131313850834 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance
Call count: 1441812

Maximum execution time: 14.3315 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.5875819762909609 ms
Execution time standard deviation: 0.33888528755720376 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance
Call count: 912676
Maximum execution time: 3.8179 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.17948720860414863 ms
Execution time standard deviation: 0.13039417460032995 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance
Call count: 591291
Maximum execution time: 15.7326 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.25391995886966373 ms
Execution time standard deviation: 0.07366556203639579 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 9.2883 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6069138443847342 ms
Execution time standard deviation: 0.3422932096018319 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance
Call count: 912743
Maximum execution time: 7.7473 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1862063900791388 ms
Execution time standard deviation: 0.13286449798014446 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance
Call count: 592953
Maximum execution time: 4.5411 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.26252990388782577 ms
Execution time standard deviation: 0.06439093796236593 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 8.6712 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6084710311052974 ms
Execution time standard deviation: 0.3429428014468045 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance
Call count: 913082
Maximum execution time: 15.1199 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1868865417344763 ms
Execution time standard deviation: 0.1346699935705888 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance
Call count: 593952
Maximum execution time: 9.3286 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.26433188826706394 ms

Execution time standard deviation: 0.06661945661942585 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 9.189 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6202202502822997 ms
Execution time standard deviation: 0.344837744109423 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance
Call count: 913826
Maximum execution time: 6.3693 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.19038073758023388 ms
Execution time standard deviation: 0.13420896776611732 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance
Call count: 593564
Maximum execution time: 4.54 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2684514310167101 ms
Execution time standard deviation: 0.06088102169488863 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance
Call count: 1441812
Maximum execution time: 16.45 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.6189073276543445 ms
Execution time standard deviation: 0.34417929004528885 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance
Call count: 913771
Maximum execution time: 6.6282 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.19072974968564813 ms
Execution time standard deviation: 0.1347241262498814 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance
Call count: 593408
Maximum execution time: 3.9985 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.26739571475274804 ms
Execution time standard deviation: 0.06119405311738127 ms